

GRIMSEY AP, Iceland

WMO: 041650

Lat: 66.544N

Lon: 18.017W

Elev: 56

StdP: 14.67

Time Zone: 0.00 (E00)

Period: 06-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	19.1	21.6	5.6	7.4	23.5	9.7	9.1	24.5	41.6	31.5	38.5	30.6	17.1	30	0.908

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.0	58.7	52.7	56.3	51.3	54.3	50.0	53.7	57.7	52.1	55.4	50.7	53.5	9.9	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
51.2	55.9	55.2	50.0	53.5	53.6	49.0	51.5	52.3	22.4	58.1	21.5	55.6	20.7	53.6	60.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 37.7	32.7	29.1	16.4	64.3	2.9	2.8	14.4	66.3	12.7	67.9	11.1	69.4	9.0	71.4		
(5)			14.1	56.3	2.6	2.5	12.2	58.1	10.7	59.6	9.2	61.0	7.3	62.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	39.1	33.6	32.9	32.5	35.3	39.2	44.9	48.1	48.1	45.3	39.6	35.4	33.8
	DBStd	7.55	4.80	5.06	5.17	5.32	4.93	4.44	3.86	3.54	4.84	5.10	5.25	5.51
	HDD50	4053	508	478	541	441	337	164	86	80	154	324	437	503
	HDD65	9455	973	898	1006	890	801	604	525	524	590	789	887	968
	CDD50	71	0	0	0	0	1	9	26	21	14	0	0	0
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	14.7	16.7	17.1	17.0	14.4	12.7	10.4	11.1	11.8	13.9	16.1	17.2	18.3
Precipitation	PrecAvg	26.5	2.5	2.3	2.4	1.5	1.3	1.5	1.7	2.0	2.6	3.0	2.9	2.5
	PrecMax	39.4	5.5	5.4	6.8	4.1	3.3	3.9	3.0	4.5	6.7	5.7	7.2	4.7
	PrecMin	16.3	0.4	0.0	0.2	0.2	0.0	0.5	0.7	0.3	0.4	0.6	1.0	0.3
	PrecStd	5.2	1.4	1.3	1.5	0.7	0.7	0.8	0.7	1.1	1.3	1.4	1.2	1.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.6	45.6	46.4	52.8	55.9	59.9	63.0	60.9	60.1	53.0	49.5	47.7
		MCWB	42.5	41.2	41.6	44.8	47.5	51.6	56.3	54.5	53.1	47.5	45.5	43.2
	2%	DB	43.8	43.3	43.0	48.0	51.9	56.2	58.8	57.4	56.2	50.3	45.7	45.0
		MCWB	40.0	39.4	39.5	43.0	46.0	50.3	53.6	53.0	51.1	45.9	42.5	41.6
	5%	DB	41.8	41.7	41.0	44.9	49.1	53.4	55.9	54.8	53.9	48.4	44.1	43.0
		MCWB	38.7	38.4	38.1	40.7	44.4	49.1	51.6	50.8	49.4	44.6	41.4	40.0
	10%	DB	39.8	40.0	39.1	42.7	46.4	51.2	53.6	53.1	52.0	46.6	42.8	41.5
		MCWB	37.4	37.2	36.3	38.9	42.8	47.6	50.4	49.5	48.2	43.8	40.4	38.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	42.8	41.9	41.9	45.8	48.0	53.5	57.1	55.3	54.0	48.0	45.6	43.8
		MCDB	46.2	44.8	45.5	51.6	53.9	57.2	62.5	59.4	59.4	51.3	48.7	46.6
	2%	WB	40.5	40.0	39.6	43.5	46.0	51.6	54.0	53.2	51.6	46.6	43.4	42.1
		MCDB	43.3	42.7	42.6	48.2	50.6	55.4	58.0	56.9	55.6	49.3	45.0	44.4
	5%	WB	39.1	38.8	38.2	41.3	44.6	49.6	52.1	51.5	50.1	45.4	42.0	40.6
		MCDB	41.3	41.2	40.7	44.7	48.4	53.0	55.2	54.2	53.1	47.7	43.6	42.6
	10%	WB	37.7	37.5	36.7	39.5	42.9	48.1	50.8	50.2	48.8	44.1	40.6	39.0
		MCDB	39.6	39.8	38.7	42.3	46.0	50.8	53.1	52.4	51.4	46.3	42.5	41.4
Mean Daily Temperature Range	5% DB	MDBR	5.7	5.7	6.2	6.5	6.6	6.7	6.2	6.0	6.3	5.5	5.9	6.0
		MCDBR	7.7	7.3	8.7	9.8	10.8	9.8	10.3	8.7	8.9	7.2	7.2	7.7
		MCWBR	6.6	6.2	7.2	6.7	6.3	6.3	6.3	5.5	5.7	5.5	6.5	6.8
	5% WB	MCDBR	7.0	6.9	8.5	9.9	9.9	8.5	9.5	8.1	8.2	6.7	6.5	7.2
		MCWBR	6.1	6.1	7.2	6.8	6.2	5.9	6.1	5.4	5.7	5.4	6.1	6.6
Clear-Sky Solar Irradiance	taub	0.369	0.296	0.318	0.347	0.350	0.343	0.345	0.340	0.329	0.299	0.361	0.374	
	taud	2.547	2.216	2.391	2.372	2.435	2.491	2.523	2.542	2.492	2.252	2.558	2.554	
	Ebn at Noon	60	173	237	258	270	275	271	260	233	167	57	10	
	Edh at Noon	6	16	22	29	30	29	28	24	20	15	5	2	
All-Sky Solar Radiation	RadAvg	17	111	382	865	1295	1574	1326	990	553	181	32	1	
	RadStd	1	14	53	87	125	153	129	90	39	21	3	0	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (7 neighbors)	+0.65	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page